

Technical Document

Project Name	Cruise	Jira ID	CCUIUXD-140
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Brief Details	Twilio Chat		

END-TO-END CHAT SYSTEM FLOW

(Twilio Conversations → S3 → Final Transcript)

PHASE 0 — MEETING SETUP

0.1 Create Video Meeting

- Backend creates Twilio Video Room
- Backend creates Twilio Conversation (Chat Channel)
- Save:
 - roomSid
 - conversationSid
- Store conversation_id in user_videocall

PHASE 1 — DURING MEETING (REAL-TIME MODE)

1.1 Participant Joins

Webhook → handleParticipantConnected

What happens:

1. Read RoomName, ParticipantIdentity
2. Extract callId, userId
3. Identify role (Recruiter / Candidate)
4. Update join/rejoin status in DB
5. Add user to Twilio Conversation

1.2 Live Chat Messaging

What happens:

- Text messages → stored only in Twilio
- File messages → stored only in Twilio Media
- Features supported:
 - Typing indicator
 - Edit / delete message
 - Participant list

1.3 Participant Leaves

Webhook → handleParticipantDisconnected

1. Update leave status
2. Remove user from chat channel
3. Prevent future messages

PHASE 2 — MEETING ENDS (EXPORT MODE)

2.1 Trigger Export

Method → generateRecord(roomSid)

What happens:

1. Fetch conversationSid
2. Fetch chat DB row (messageId)
3. Start full export

PHASE 3 — FILE EXPORT

3.1 Fetch All Messages from Twilio

```
Message.reader(SERVICE_SID, conversationSid).read();
```

3.2 Identify Real Media

For each message:

- Only type = media
- Skip:

- text/plain
- already processed media

3.3 Insert PROCESSING Record

Table: videocall_chat_files

Status:

- 13 → PROCESSING

Purpose:

- Track partial progress
- Enable retries
- Avoid duplicates

3.4 Download Media from Twilio

- Fetch Media metadata (MCS API)
- Generate secure temporary URL
- Download bytes using Basic Auth

3.5 Upload Media to AWS S3

- Path:

chatfiles/{conversationSid}/{timestamp}_{mediaSid}

3.6 Update Status to COMPLETED

Status:

- 3 → COMPLETED
- Save S3 URL

If failed:

- 14 → FAILED

PHASE 4 — TRANSCRIPT JSON GENERATION

4.1 Build Final JSON

Method → generateChatJson()

What happens:

1. Fetch all messages again
2. Convert:
 - a. Text → text JSON
 - b. Media → media JSON (with S3 URL)
3. Merge into ordered list

4.2 Upload JSON to S3

- Path:

records/{roomId}_{timestamp}.json

4.3 Save JSON URL in DB

Table: videocall_chat_messages

PHASE 5 — RETRY SCHEDULER (SELF-HEALING)

5.1 Retry Job Runs

Method → retryMediaUploads()

What happens:

- Twilio media count
- DB uploaded media count

5.2 Retry Missing Media

For each missing media:

1. Download again from Twilio
2. Upload to S3
3. Update DB

PHASE 6 — CLEANUP SCHEDULER (COST + SECURITY)

6.1 Cleanup Job Runs

Method → cleanupConversations()

What happens:

- Twilio media count == DB media count
- Media count > 0

6.2 Delete Twilio Conversation

```
Channel.deleter(SERVICE_SID, conversationSid).delete();
```

Then:

- Mark deleted in DB

ONE-LINE SUMMARY

Chat runs live on Twilio, after the meeting we export all messages and files to S3, build one final JSON transcript, auto-retry failures, and delete Twilio data only after verification.